# Yifan Feng

Ph.D. Candidate in Entomology — mosquito genomics, chromosomal inversions, and cytogenetics

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## Education

**PhD** **Virginia Polytechnic Institute and State University**, Entomology — Blacksburg, VA — Aug 2021 – Dec 2026

- Advisor: Dr. Maria V. Sharakhova
- Dissertation: *Chromosomal Inversions Differentiate Mosquitoes in the Culex pipiens Complex*
- Department of Entomology and Fralin Life Sciences Institute
- Coursework: Molecular Genetics, Bioinformatics, Genomics, Biochemistry, Experimental Design

**BS** **Virginia Polytechnic Institute and State University**, Biological Sciences — Blacksburg, VA — Aug 2017 – May 2021

**BS** **Virginia Polytechnic Institute and State University**, Food Science and Technology — Blacksburg, VA — Aug 2017 – May 2021

## Research Experience

**Virginia Tech — Department of Entomology & Fralin Life Sciences Institute**, Graduate Research Assistant — Blacksburg, VA — Aug 2021 – present

- Led multi-species genomic projects integrating **Hi-C scaffolding**, **Nanopore long-read sequencing**, and **FISH cytogenetic mapping** to assemble chromosome-scale mosquito genomes.
- Designed and validated **PCR breakpoint assays** to genotype chromosomal inversions across natural populations, linking structural variants to climatic adaptation and vector traits.
- Conducted population genomic analyses (**FST, PCA, AMOVA**) to quantify differentiation and gene flow among *Culex pipiens* populations across North America.
- Performed molecular dissections, DNA extraction, library preparation, and fluorescence microscopy imaging; maintained mosquito colonies.
- Mentored two undergraduate researchers in PCR optimization, experimental design, and scientific writing.
- Coordinated with collaborating groups across nine countries on the *Aedes aegypti* inversion study published in *Genome Biology and Evolution*.

**Virginia Tech — Department of Food Science and Technology**, Undergraduate Researcher — Blacksburg, VA — Sept 2020 – Dec 2020

Aerosolized *Listeria* contamination from cavitation treatment of raw produce.

- Applied microbubble cavitation treatments and quantified resulting aerosolized bacterial contamination.
- Built MATLAB models of cavitation effects and ran predictive simulations on university high-performance computing resources.
- Compiled, visualized, and statistically analyzed experimental results to identify contamination trends.

**Molecular Genetics Laboratory**, Research Intern — Blacksburg, VA — Mar 2020 – Dec 2021

- Performed DNA extraction, PCR amplification, and sequencing for mosquito population genetics studies.
- Managed sample inventory and quality control for field and laboratory collections.
- Gained hands-on experience with genome assembly workflows and variant calling pipelines.

## Publications

**Chromosomal Inversions and Their Potential Impact on the Evolution of Arboviral Vector *Aedes aegypti*** — July 2025
Jiangtao Liang, Noah H. Rose, Ilya I. Brusentsov, Varvara Lukyanchikova, Dmitriy A. Karagodin, *Yifan Feng*, et al.
10.1093/gbe/evaf118 (Genome Biology and Evolution 17(7), evaf118)

**Hi-C Proximity Ligation Approach Identified Chromosomal Rearrangements in *Culex pipiens* Mosquitoes** — 2025
*Yifan Feng*, Varvara Lukyanchikova, Jiangtao Liang, Dmitriy A. Karagodin, Ilya I. Brusentsov, Megan L. Fritz, et al.
(American Journal of Tropical Medicine and Hygiene 112(6) — ASTMH Annual Meeting abstract)

**Chromosomal Rearrangements Differentiate Species of the *Culex pipiens* Complex** — 2025
Maria V. Sharakhova, *Yifan Feng*, Varvara Lukyanchikova, Jiangtao Liang, Dmitriy A. Karagodin, et al.
(Conference proceedings)

**Chromosomal Inversions Differentiate Mosquitoes in the *Culex pipiens* Complex** — 2026
*Yifan Feng*, Varvara Lukyanchikova, Jiangtao Liang, Ilya I. Brusentsov, Dmitriy A. Karagodin, Megan L. Fritz, Maria V. Sharakhova
(Manuscript in preparation)

**Chromosome-Scale Genome Assembly of *Aedes albopictus*** — 2026
*Yifan Feng*, et al.
(Manuscript in preparation)

## Conference Presentations

- **Poster** — *Hi-C based discovery of structural variants in the Culex pipiens complex.* American Society of Tropical Medicine and Hygiene (ASTMH) Annual Meeting, November 2024.
- **Presentation** — *Chromosomal inversions differentiate mosquitoes in the Culex pipiens complex.* CeZAP Infectious Disease Symposium, Blacksburg, VA, October 2023.

## Teaching and Mentoring

**Virginia Tech — Department of Entomology**, Graduate Teaching Assistant, ENT 2004 (Insects and Human Society) — Blacksburg, VA — Aug 2023 – May 2024

- Led weekly laboratory sessions and discussion sections for a general-education course serving non-biology majors.
- Taught insect taxonomy, physiology, and ecology exercises; graded laboratory reports and examinations.
- Developed visual teaching materials and provided one-on-one mentorship.

**Sharakhova Lab, Virginia Tech**, Research Mentor, Undergraduate Researchers — Blacksburg, VA — Jan 2024 – Dec 2025

- Mentored two undergraduate researchers through full project cycles — PCR optimization and troubleshooting, experimental design, data interpretation, and scientific writing.

## Technical Skills

**Molecular Biology:** DNA/RNA extraction, PCR, qPCR, RT-qPCR, ddPCR, gel electrophoresis, cloning, nucleic acid purification

**Library Prep & Sequencing:** Hi-C library preparation, Illumina library prep, Sanger sequencing, Nanopore long-read workflows

**Cytogenetics & Imaging:** Chromosome preparation, FISH probe synthesis and hybridization, fluorescence microscopy (Zeiss ZEN), ImageJ

**Genomics & Bioinformatics:** Hi-C scaffolding (Juicer, Juicebox), genome assembly, trio-binning, gene annotation, variant calling, alignment (BWA, Minimap2)

**Population Genetics:** FST, PCA, AMOVA, STRUCTURE, PLINK, inversion genotyping

**Cell & Assay Work:** Mammalian and insect cell culture, sterile technique, ELISA, Western blot, BSL-2 practice

**Programming & Software:** R (ggplot2), Python, Linux/Unix, MATLAB, Galaxy, Geneious, BLAST/NCBI, GraphPad Prism, BioRender

**Scientific Communication:** Manuscript preparation, grant proposal writing, poster and oral presentation, mentoring, laboratory documentation

## Service and Professional Membership

- Volunteer, Blacksburg Mosquito Control Initiative — field collection and public outreach on vector-borne disease prevention.
- Member, Entomological Society of America.
- Member, Virginia Mosquito Control Association.